



Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Confusion Matrix, Precision, and Recall

WEEK 2 · DAY 2

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

A probability becomes a flag

Day 6

a model probability of default in this file

Today

a threshold turns it into a flag, 1 or 0

Then

count the flags against what happened

Real data, one place and year

30,000 card accounts, Taiwan, 2005. UCI, CC BY 4.0. Credit there differs from US credit.

Never model features

SEX, EDUCATION, MARRIAGE, AGE. Every week 2 model reads the other 19 columns.

18,000 training: fit

6,000 validation

6,000 test: day 10

Default rate 22.12 percent in every part. Today: training and validation rows only.

What the output is called

Allowed names

model probability of default in this file

flagged at threshold t

the model's score

Never called

a credit decision

an approval, a decline

a lending policy

a risk grade for a real person

The file describes accounts at one lender, in one country, in one year. A model fitted to it describes that file.

The all-zero model and its accuracy

A model that flags nobody

0.7788

accuracy on the validation rows

0 of 1,327

defaults flagged

Day 6's model, flagged at 0.5

0.8123

accuracy on the validation rows

324 of 1,327

defaults flagged

The all-zero model is right on every account that paid, 4,673 of 6,000, because they are most of the rows. On a file where most rows are 0, accuracy mostly measures how many rows are 0.

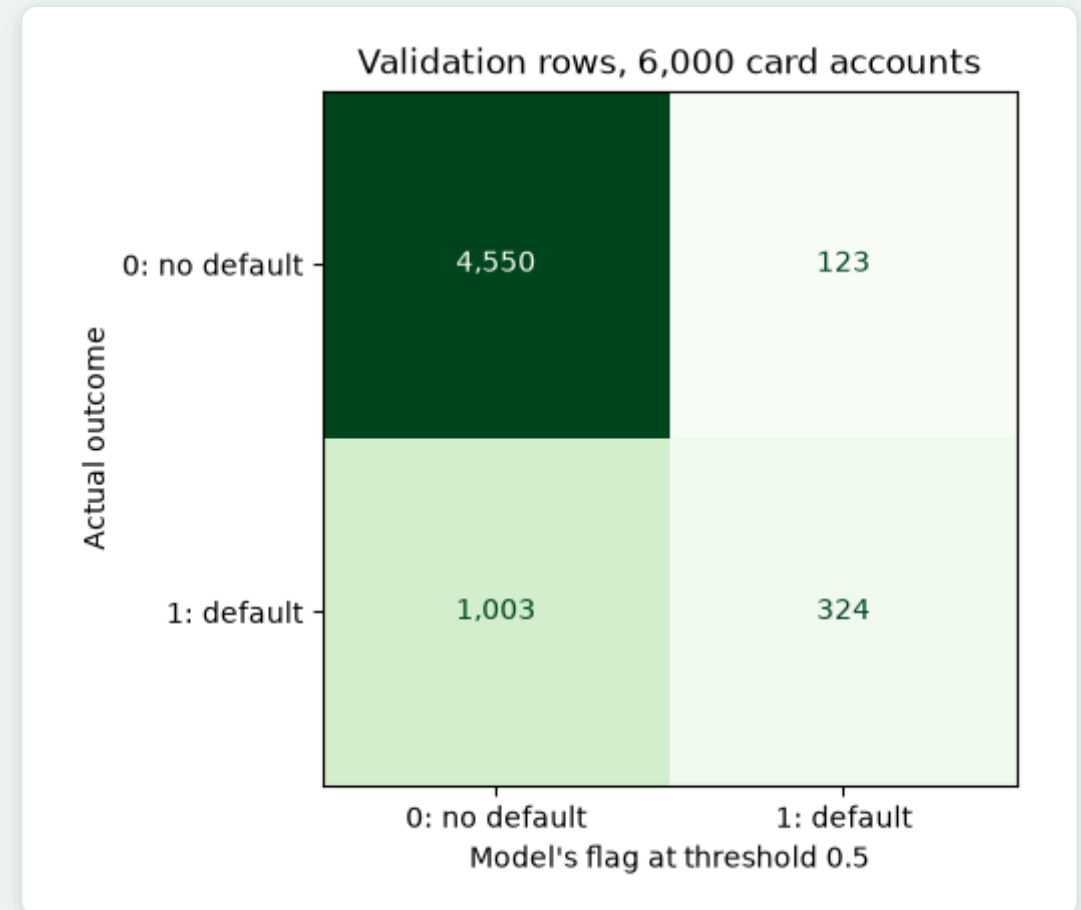
Accuracy is always shown beside the all-zero model's, and never used to choose.

The confusion matrix, cell by cell

	Flagged 0	Flagged 1
Actual 0: paid	TN 4,550	FP 123
Actual 1: default	FN 1,003	TP 324

Rows: 4,673 paid, 1,327 defaulted. Columns: 447 flagged, 5,553 not.

FN = 1,003: the defaults accuracy hid.



Precision, recall, and F1, for the defaults

Precision

$$TP / (TP + FP)$$

Of the accounts flagged, the share that defaulted

0.7248

Recall

$$TP / (TP + FN)$$

Of the defaults, the share flagged

0.2442

F1

$$2TP / (2TP + FP + FN)$$

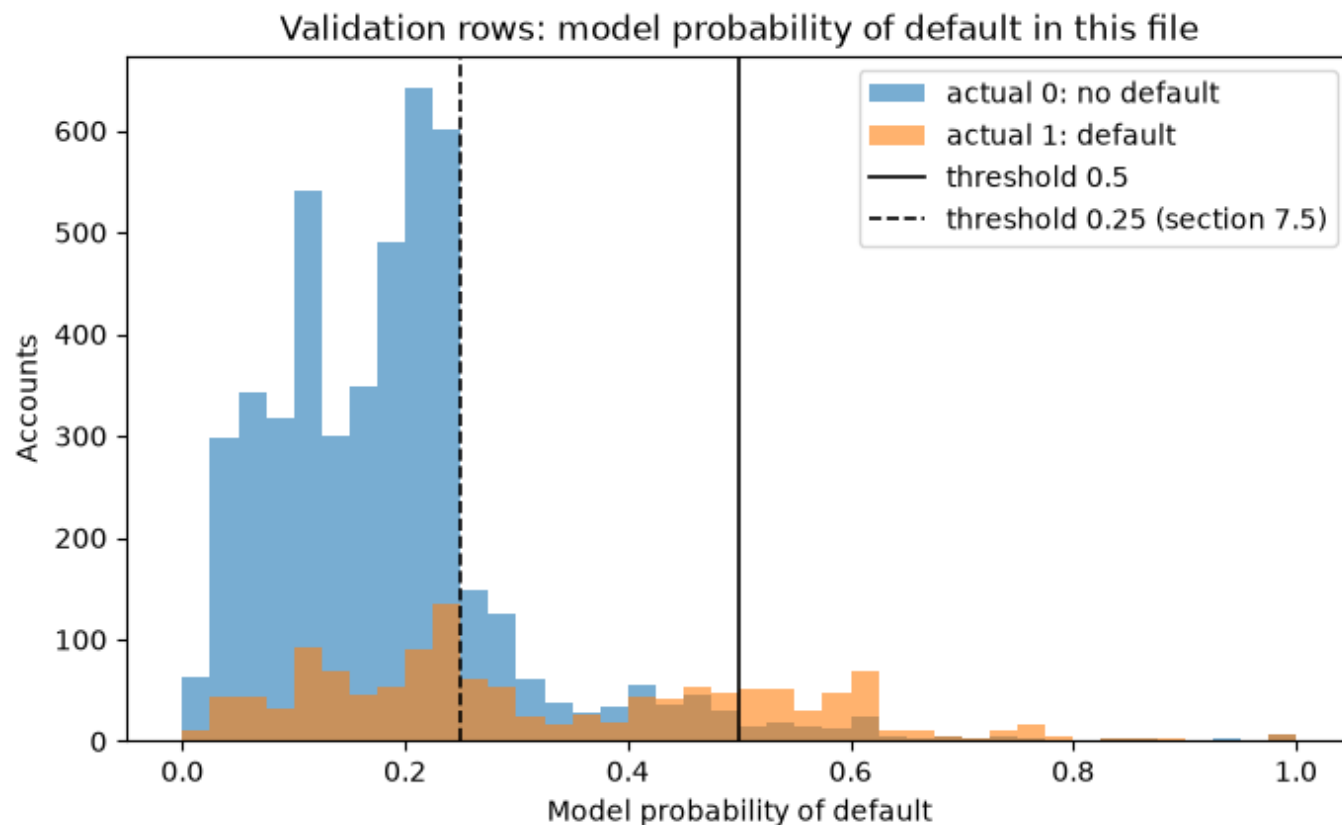
Harmonic mean: high only when both are

0.3653

Precision reads down the flagged 1 column. Recall reads along the actual 1 row. Neither uses the true negatives, so neither is flattered by a file full of zeros.

At 0.5: a flag is a default about 7 times in 10, and about 3 defaults in 4 go unflagged. 31 tests pass.

Threshold 0.5 and what it catches



447

accounts flagged, 7.45% of the rows

324 of 1,327

defaults flagged (recall 0.2442)

predict against $\text{proba} \geq t$

predict flags above 0.5; the course flags at least t .
Here: the same flags.

What each error costs, as arithmetic on counts

False positive

An account that paid, flagged: attention spent on it.

False negative

A default, not flagged: the flag did not point to it.

Threshold	TP	FP	FN	Recall	Precision	Accuracy
0.5	324	123	1003	0.2442	0.7248	0.8123
0.25	710	727	617	0.5350	0.4941	0.7760

1.56

false flags per extra
default

From 0.5 to 0.25: 386 more defaults caught, 604 more false flags. Counting a miss as w false flags, the smaller total is at 0.5 for $w = 1$ and at 0.25 for $w = 2$.

The weight is a choice the file cannot make. No count here is a rule a lender uses.

classification_report, read by row

	precision	recall	f1-score	support
0	0.8194	0.9737	0.8899	4673
1	0.7248	0.2442	0.3653	1327
accuracy			0.8123	6000
macro avg	0.7721	0.6089	0.6276	6000
weighted avg	0.7985	0.8123	0.7739	6000

- Row 1

the defaults: the numbers of 7.3
- Row 0

0 as positive: recall 0.9737 is not the defaults'
- macro avg

plain average of the two rows
- weighted avg

leans toward class 0, the larger

The course reads row 1. A summary quoting recall 0.9737 for this model read the wrong row.

Excel side by side: four COUNTIFS

Excel side by side (tested in Excel)

F5 =COUNTIFS(\$A\$2:\$A\$6001,1,\$B\$2:\$B\$6001,">="&F\$1)

F4 =COUNTIFS(\$A\$2:\$A\$6001,1,\$B\$2:\$B\$6001,"<"&F\$1)

F3 =COUNTIFS(\$A\$2:\$A\$6001,0,\$B\$2:\$B\$6001,">="&F\$1)

F2 =COUNTIFS(\$A\$2:\$A\$6001,0,\$B\$2:\$B\$6001,"<"&F\$1)

=F5/(F5+F3) and =F5/(F5+F4)

=2*F5/(2*F5+F3+F4)

=COUNTIF(\$A\$2:\$A\$6001,0)/COUNT(\$A\$2:\$A\$6001)

TP 324

FN 1,003

FP 123

TN 4,550

precision 0.7248, recall 0.2442

F1 0.3653

all-zero accuracy 0.7788

Sheet valid: A the outcome, B the model's probability, threshold in F1. Typing 0.25 there recounts everything: TP 710, FP 727.

Counts equal Python's exactly; every ratio within 1e-12.

Git: a change of threshold in the log

1. Code

mlkit.py and test_mlkit.py

2. Log at 0.5

results.csv, 4 decimals

3. Log at 0.25

the same model, threshold moved

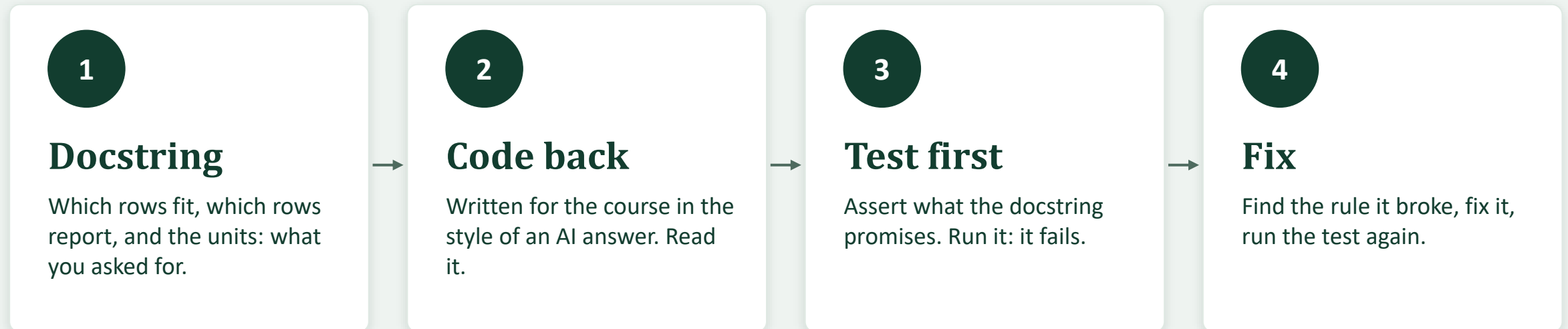
```
!git -C "{work}" diff --word-diff-regex="^[[:space:]]+" results.csv
logistic_19_columns_scaled,19_non_person_columns,[-0.5000-]{+0.2500+},18000,6000,
[-0.8123,0.7248,0.2442,0.3653-]{+0.7760,0.4941,0.5350,0.5137+}
```

Columns: threshold, train_rows, valid_rows, valid_accuracy, valid_precision, valid_recall, valid_f1. Only the threshold and the four metrics are marked.

Plain --word-diff splits at spaces; a CSV row has none, so tell Git a word ends at a comma.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.



Today: recall computed with `pos_label=0`: 0.9737, the accounts that paid, against 0.2442 for the defaults.

Code that runs is not the same as code that is right.

What you can do now

- Flagging nobody scores 0.7788: accuracy hides the rare class
- Four counts: TN, FP, FN, TP
- Precision down the flagged column, recall along the default row
- A lower threshold trades false flags for caught defaults
- In `classification_report`, the defaults are row 1

Week 2, Day 3 ROC curves and choosing a threshold on validation rows